

What does $\nabla \times \vec{B}$ really mean?

From magnetostatics, we have the differential form of Ampere's law:

$$\nabla \times \vec{B} = \mu_0 \vec{J}. \quad (1)$$

This is a very important equation, but we can often ask a natural question:

$$\text{If magnetic field lines are circular around a current, why is } \nabla \times \vec{B} \text{ not nonzero everywhere?} \quad (2)$$

To answer this properly, we must distinguish between two different ideas:

- the **global shape** of magnetic field lines, and
- the **local circulation per unit area**, which is what curl measures.

Key Idea

A field may have **curved** or even **circular-looking** field lines and still have zero curl at a point. Curl is not a test for whether field lines look bent. Curl is a test for whether there is a **local source of circulation** at that point.

Curl as circulation per unit area

From Stokes' theorem, the component of curl along the unit normal $\hat{n}$ is given by

$$(\nabla \times \vec{B}) \cdot \hat{n} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \vec{B} \cdot d\vec{l} \quad (3)$$

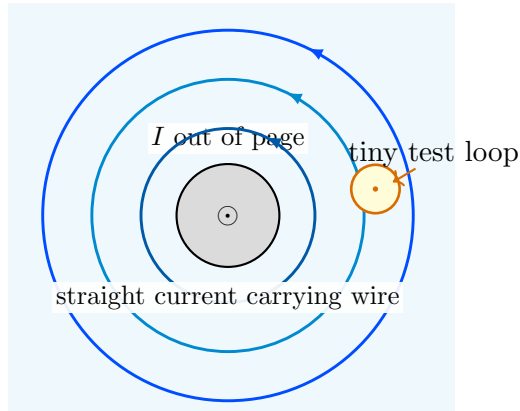
where C is a very small closed loop bounding the small surface element ΔS .

So curl answers the following local question:

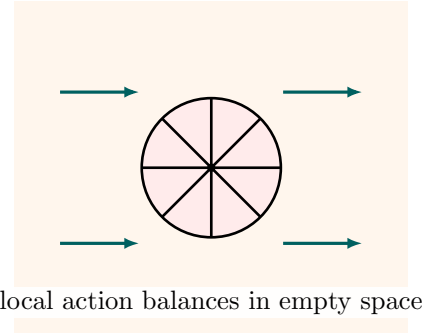
If we place an infinitesimally small loop around a point, does the magnetic field produce a nonzero circulation per unit area around that point?

A useful physical picture is a tiny paddle wheel. If the field tends to rotate that tiny wheel, then the curl is nonzero. If the local action balances, then the curl is zero.





(a) The magnetic field lines are circular around the wire.



(b) Curl depends on local imbalance, not merely on the shape of field lines.

Figure 1: A logical way to think about curl.

Case 1: Long straight current carrying conductor

Let a long straight conductor carry a steady current I along the z -direction. The magnetic field is

$$\vec{B} = \hat{\phi} \frac{\mu_0 I}{2\pi\rho}. \quad (4)$$

Thus, the magnetic field lines are concentric circles around the conductor.

Outside the conductor

At any point outside the conductor, the current density is zero:

$$\vec{J} = 0. \quad (5)$$

Therefore, from (1),

$$\boxed{\nabla \times \vec{B} = 0 \quad \text{outside the conductor.}} \quad (6)$$

This is true even though the field lines are circular. The field exists there, and it bends there, but there is no **local current density** at that point.

Inside the conductor

Now assume the wire has finite radius a and carries a uniform current density $\vec{J} = J\hat{z}$. Then inside the conductor,

$$B_\phi(\rho) = \frac{\mu_0 J \rho}{2}, \quad 0 \leq \rho \leq a. \quad (7)$$

Since current actually flows through that region,

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J} \neq 0 \quad \text{inside the conductor.}} \quad (8)$$



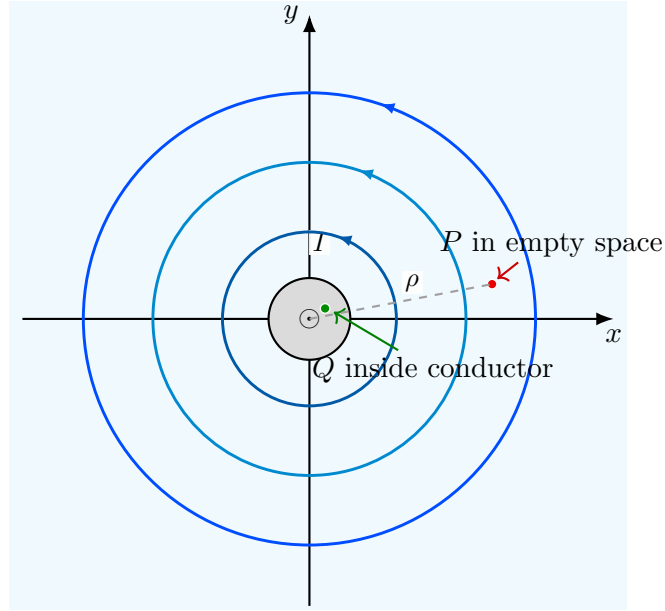


Figure 2: Straight wire carrying current along $+z$. Point P lies outside the conductor, whereas point Q lies inside the conducting region.

Important Clarification

The phrase “near the wire” is not precise enough. A point may be very close to the conductor and still lie in empty space. Curl becomes nonzero not because the point is merely *near* the current, but because the point lies in the region where **current density exists**.

Why there is no contradiction with Ampere’s law

We often compare the above result with the integral form of Ampere’s law:

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}. \quad (9)$$

For a circular loop around the wire, this integral is clearly nonzero. Then why is the curl zero outside the conductor?

The answer is that (9) is a statement about a **finite closed loop**. Such a loop may enclose the conductor and hence enclose current. But curl is a **local** concept. To test the curl at a point, we shrink the loop around that point.

- If the point lies outside the conductor, an infinitesimal loop around it encloses no current.
- If the point lies inside the conductor, an infinitesimal loop around it encloses current.

So there is no contradiction at all.



Key Takeaways

- A **finite** Amperian loop around the wire can enclose current and give nonzero circulation.
- An **infinitesimal** loop around a point outside the conductor encloses no current, so the local curl there is zero.
- Thus, field lines may look circular globally, while the local curl can still be zero at most nearby points.

Case 2: Small current carrying loop

Now consider a small circular loop carrying steady current.

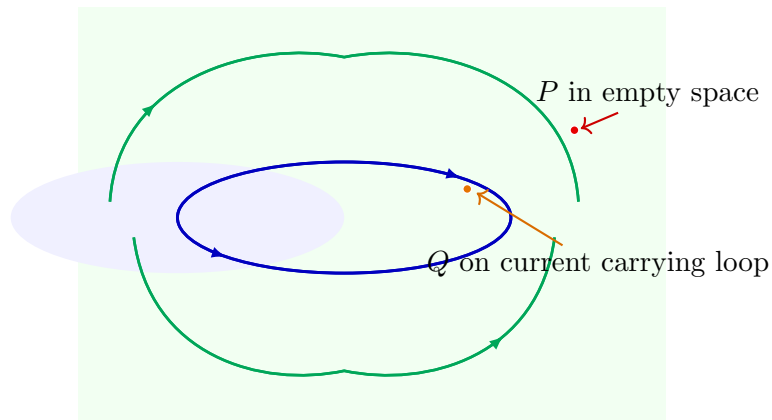


Figure 3: A small current loop produces a magnetic field in the surrounding space. However, the curl of $\vec{B}$ is nonzero only where current actually flows.

In the empty space around the loop, there is no current density. Therefore,

$$\vec{J} = 0 \quad \Rightarrow \quad \nabla \times \vec{B} = 0. \quad (10)$$

This remains true even if the field is strong, curved, or varying rapidly with position. Strong variation does **not** automatically imply nonzero curl.

Only in the conducting material of the loop, where current actually flows, do we have

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J}.} \quad (11)$$

A very clean local test

The cleanest way to explain the result is the following:

$$\boxed{\nabla \times \vec{B} \neq 0 \text{ at a point} \iff \text{an infinitesimal loop around that point encloses current.}} \quad (12)$$

So, for the two examples:

- **Straight conductor:**

- outside conductor $\Rightarrow$ no current enclosed locally $\Rightarrow$ curl zero,
- inside conductor $\Rightarrow$ current enclosed locally $\Rightarrow$ curl nonzero.

- **Small current loop:**

- point in empty space near the loop $\Rightarrow$ no local current enclosed $\Rightarrow$ curl zero,
- point in the conducting wire itself $\Rightarrow$ current enclosed locally $\Rightarrow$ curl nonzero.

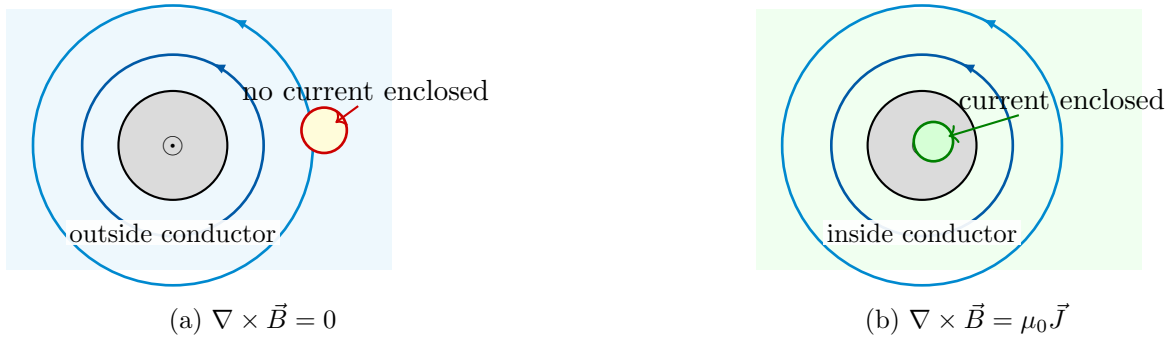


Figure 4: What matters is not mere closeness to the wire, but whether current passes through the infinitesimal loop around the point.

Connection with the ideal thin wire

For an ideal infinitely thin wire, all the current is mathematically concentrated on the wire axis. Then the current density is singular there. In that idealized case:

- the curl is zero everywhere away from the axis,
- the nonzero contribution is concentrated on the axis itself in the distributional sense.

So in practice, when we say that curl is nonzero “at the wire”, we really mean that for a real conductor the nonzero curl exists in the finite conducting region carrying current.

Final boxed summary

Boxed Summary

- Magnetostatics gives

$$\nabla \times \vec{B} = \mu_0 \vec{J}.$$

- Curl measures **local circulation per unit area**, not merely the visual curvature of field lines.
- A magnetic field may have circular field lines and still have

$$\nabla \times \vec{B} = 0$$

at a point if no current density is present there.

- For a long straight conductor,

$$\nabla \times \vec{B} = 0 \quad \text{outside the conductor,} \quad \nabla \times \vec{B} = \mu_0 \vec{J} \quad \text{inside the conductor.}$$

- For a small current loop, the magnetic field fills the surrounding space, but the curl is nonzero only in the conducting material of the loop.
- Therefore, the correct statement is:

Curl of $\vec{B}$ is nonzero exactly where current density exists.